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Exp.2 Inter-VLAN Routing

Objectives

1. Describe options for configuring inter-VLAN routing.
2. Configure legacy inter-VLAN routing.
3. Configure router-on-a-Stick Inter-VLAN routing.
4. Configure inter-VLAN routing using Layer 3 switching.

Introduction

Now that you know how to configure VLANs on a network switch, the next step is to allow devices connected to the various VLANs to communicate with each other. In a previous experiment, you learned that each VLAN is a unique broadcast domain, so computers on separate VLANs are, by default, not able to communicate. There is a way to permit these end stations to communicate; it is called inter-VLAN routing. In this experiment, you will learn what inter-VLAN routing is and some of the different ways to accomplish inter-VLAN routing on a network.

There are three inter-VLAN routing options:

- **Legacy Inter-VLAN routing** - This is a legacy solution. It does not scale well.
- **Router-on-a-Stick** - This is an acceptable solution for a small to medium-sized network.
- **Layer 3 switch using switched virtual interfaces (SVIs)** - This is the most scalable solution for medium to large organizations.

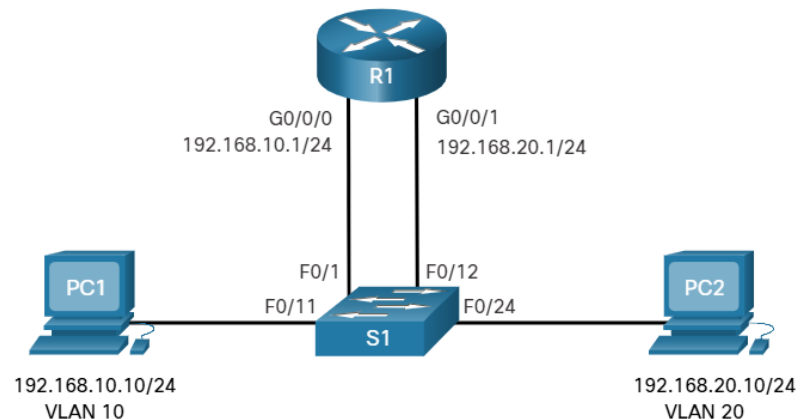
Legacy Inter-VLAN routing

The first inter-VLAN routing solution relied on using a router with multiple Ethernet interfaces. Each router interface was connected to a switch port in different VLANs. The router interfaces served as the default gateways to the local hosts on the VLAN subnet.

For example, refer to the topology where R1 has two interfaces connected to switch S1.

Notice in the example MAC address table of S1 is populated as follows:

- Fa0/1 port is assigned to VLAN 10 and is connected to the R1 G0/0/0 interface.
- Fa0/11 port is assigned to VLAN 10 and is connected to PC1.
- Fa0/12 port is assigned to VLAN 20 and is connected to the R1 G0/0/1 interface.
- Fa0/24 port is assigned to VLAN 20 and is connected to PC2.



When PC1 sends a packet to PC2 on another network, it forwards it to its default gateway 192.168.10.1. R1 receives the packet on its G0/0/0 interface and examines the destination address

of the packet. R1 then routes the packet out its G0/0/1 interface to the F0/12 port in VLAN 20 on S1. Finally, S1 forwards the frame to PC2.

Legacy inter-VLAN routing using physical interfaces works, but it has a significant limitation. It is not reasonably scalable because routers have a limited number of physical interfaces. Requiring one physical router interface per VLAN quickly exhausts the physical interface capacity of a router.

In our example, R1 required two separate Ethernet interfaces to route between VLAN 10 and VLAN 20. What if there were six (or more) VLANs to interconnect? A separate interface would be required for each VLAN. Obviously, this solution is not scalable.

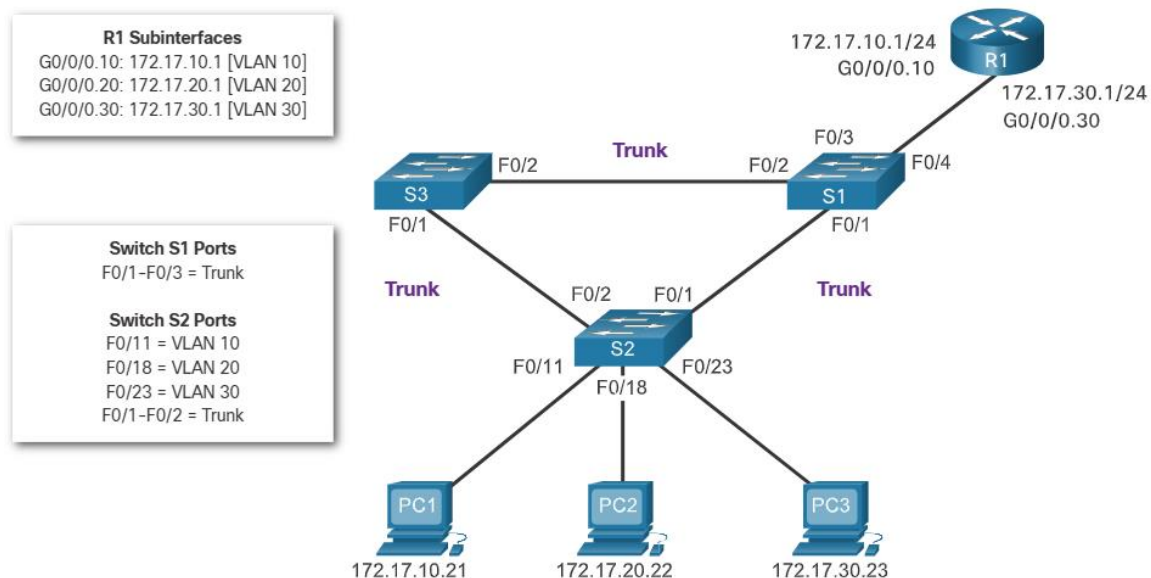
Router-on-a-Stick Inter-VLAN Routing

The 'router-on-a-stick' inter-VLAN routing method overcomes the limitation of the legacy inter-VLAN routing method. It only requires one physical Ethernet interface to route traffic between multiple VLANs on a network.

A Cisco IOS router Ethernet interface is configured as an 802.1Q trunk and connected to a trunk port on a Layer 2 switch. Specifically, the router interface is configured using subinterfaces to identify routable VLANs.

The configured subinterfaces are software-based virtual interfaces. Each is associated with a single physical Ethernet interface. Subinterfaces are configured in software on a router. Each subinterface is independently configured with an IP address and VLAN assignment. Subinterfaces are configured for different subnets that correspond to their VLAN assignment. This facilitates logical routing.

When VLAN-tagged traffic enters the router interface, it is forwarded to the VLAN subinterface. After a routing decision is made based on the destination IP network address, the router determines the exit interface for the traffic. If the exit interface is configured as an 802.Q subinterface, the data frames are VLAN-tagged with the new VLAN and sent back out the physical interface.



As seen in the animation, PC1 on VLAN 10 is communicating with PC3 on VLAN 30. When R1 accepts the tagged unicast traffic on VLAN 10, it routes that traffic to VLAN 30, using its

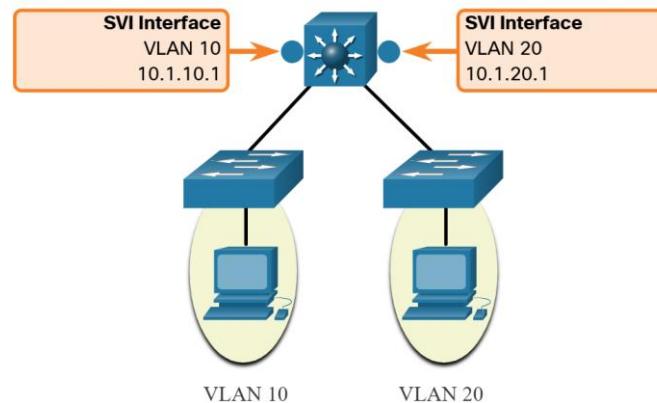
configured subinterfaces. Switch S2 removes the VLAN tag of the unicast frame and forwards the frame out to PC3 on port F0/23.

Note: The router-on-a-stick method of inter-VLAN routing does not scale beyond 50 VLANs.

Inter-VLAN Routing on a Layer 3 Switch

The modern method of performing inter-VLAN routing is to use Layer 3 switches and switched virtual interfaces (SVI). An SVI is a virtual interface that is configured on a Layer 3 switch, as shown in the figure.

Note: A Layer 3 switch is also called a multilayer switch as it operates at Layer 2 and Layer 3.



Inter-VLAN SVIs are created the same way that the management VLAN interface is configured. The SVI is created for a VLAN that exists on the switch. Although virtual, the SVI performs the same functions for the VLAN as a router interface would. Specifically, it provides Layer 3 processing for packets that are sent to or from all switch ports associated with that VLAN.

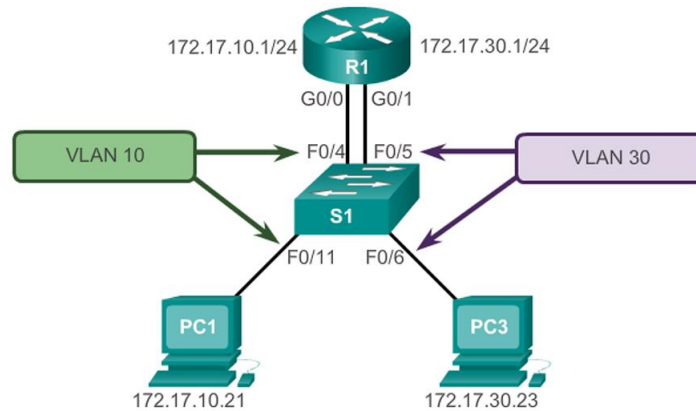
The following are advantages of using Layer 3 switches for inter-VLAN routing:

- They are much faster than router-on-a-stick because everything is hardware switched and routed.
- There is no need for external links from the switch to the router for routing.
- They are not limited to one link because Layer 2 EtherChannels can be used as trunk links between the switches to increase bandwidth.
- Latency is much lower because data does not need to leave the switch in order to be routed to a different network.
- They are more commonly deployed in a campus LAN than routers.

The only disadvantage is that Layer 3 switches are more expensive.

Legacy Inter-VLAN Routing Configuration

Legacy inter-VLAN routing requires a router to have multiple physical interfaces, with each interface connected to a unique VLAN. Every interface is assigned an IP address corresponding to the subnet of its VLAN. Devices within each VLAN use the router as their default gateway to communicate with devices in other VLANs.



Switch Configuration:

```

S1(config)# vlan 10
S1(config-vlan)# vlan 30
S1(config-vlan)# interface f0/11
S1(config-if)# switchport access vlan 10
S1(config-if)# interface f0/4
S1(config-if)# switchport access vlan 10
S1(config-if)# interface f0/6
S1(config-if)# switchport access vlan 30
S1(config-if)# interface f0/5
S1(config-if)# switchport access vlan 30
S1(config-if)# end
*Mar 20 01:22:56.751: %SYS-5-CONFIG_I: Configured from console by
console
S1# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]

```

Router Interface Configuration:

```

R1(config)# interface g0/0
R1(config-if)# ip address 172.17.10.1 255.255.255.0
R1(config-if)# no shutdown
*Mar 20 01:42:12.951: %LINK-3-UPDOWN: Interface GigabitEthernet0/0,
changed state to up
*Mar 20 01:42:13.951: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0, changed state to up
R1(config-if)# interface g0/1
R1(config-if)# ip address 172.17.30.1 255.255.255.0
R1(config-if)# no shutdown
*Mar 20 01:42:54.951: %LINK-3-UPDOWN: Interface GigabitEthernet0/1,
changed state to up
*Mar 20 01:42:55.951: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to up
R1(config-if)# end
R1# copy running-config startup-config

```

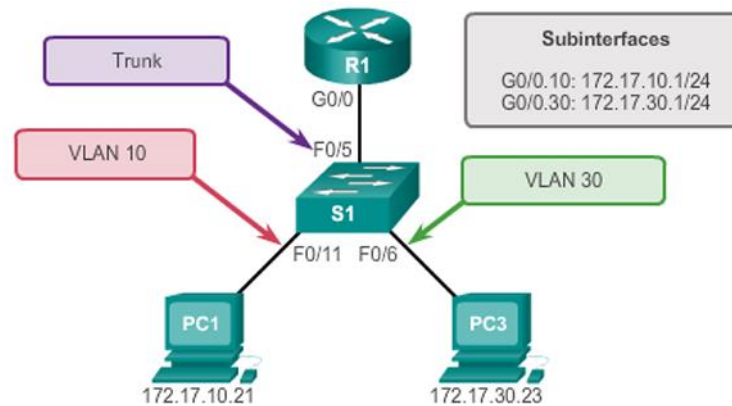
تیسری سیکٹر PC1
+ بیوت ال default gateway

تیسری سیکٹر PC3

Router-on-a-Stick Configuration

An alternative to legacy inter-VLAN routing is the use of VLAN trunking and subinterfaces. VLAN trunking enables a single physical router interface to handle traffic for multiple VLANs. The router's physical interface is connected to a trunk link on the adjacent switch, and subinterfaces are

then created on the router for each VLAN. Each subinterface is assigned an IP address corresponding to its VLAN's subnet and is configured to tag frames for that VLAN.



Switch Configuration:

```
S1(config)# vlan 10
S1(config-vlan)# vlan 30
S1(config-vlan)# interface f0/5
S1(config-if)# switchport mode trunk
S1(config-if)# end
S1#
```

Router Subinterface Configuration:

```
R1(config)# interface g0/0.10
R1(config-subif)# encapsulation dot1q 10
R1(config-subif)# ip address 172.17.10.1 255.255.255.0
R1(config-subif)# interface g0/0.30
R1(config-subif)# encapsulation dot1q 30
R1(config-subif)# ip address 172.17.30.1 255.255.255.0
R1(config)# interface g0/0
R1(config-if)# no shutdown

*Mar 20 00:20:59.299: %LINK-3-UPDOWN: Interface GigabitEthernet0/0,
changed state to down
*Mar 20 00:21:02.919: %LINK-3-UPDOWN: Interface GigabitEthernet0/0,
changed state to up
*Mar 20 00:21:03.919: %LINEPROTO-5-UPDOWN: Line protocol on
changed state to down
*Mar 20 00:21:02.919: %LINK-3-UPDOWN: Interface GigabitEthernet0/0,
changed state to up
*Mar 20 00:21:03.919: %LINEPROTO-5-UPDOWN: Line protocol on
Interface GigabitEthernet0/0, changed state to up
```

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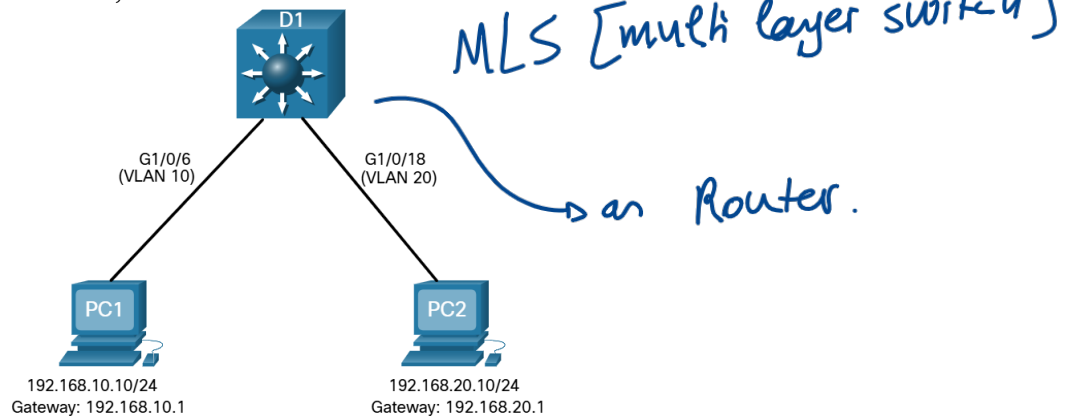
also default gateway.

Layer 3 Switch Configuration

Inter-VLAN routing using the router-on-a-stick method is simple to implement for a small to medium-sized organization. However, a large enterprise requires a faster, much more scalable method to provide inter-VLAN routing.

To provide inter-VLAN routing, Layer 3 switches use SVIs. SVIs are configured using the same **interface vlan** *vlan-id* command used to create the management SVI on a Layer 2 switch. A Layer 3 SVI must be created for each of the routable VLANs.

In the figure, the Layer 3 switch, D1, is connected to two hosts on different VLANs. PC1 is in VLAN 10 and PC2 is in VLAN 20, as shown.



Complete the following steps to configure S1 with VLANs and trunking:

- Step 1.** Create the VLANs.
- Step 2.** Create the SVI VLAN interfaces.
- Step 3.** Configure access ports.
- Step 4.** Enable IP routing.

1. Create the VLANs.

First, create the two VLANs as shown in the output.

```

D1(config)# vlan 10
D1(config-vlan)# name LAN10
D1(config-vlan)# vlan 20
D1(config-vlan)# name LAN20
D1(config-vlan)# exit
D1(config)#
  
```

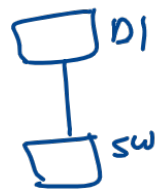
2. Create the SVI VLAN interfaces.

Configure the SVI for VLANs 10 and 20. The IP addresses that are configured will serve as the default gateways to the hosts in the respective VLANs. Notice the informational messages showing the line protocol on both SVIs changed to up.

```

D1(config)# interface vlan 10
D1(config-if)# description Default Gateway SVI for 192.168.10.0/24
D1(config-if)# ip add 192.168.10.1 255.255.255.0
D1(config-if)# no shut
D1(config-if)# exit
D1(config)#
D1(config)# int vlan 20
D1(config-if)# description Default Gateway SVI for 192.168.20.0/24
D1(config-if)# ip add 192.168.20.1 255.255.255.0
D1(config-if)# no shut
D1(config-if)# exit
D1(config)#
*Sep 17 13:52:16.053: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
*Sep 17 13:52:16.160: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
  
```

→ default gateway!



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3. Configure access ports.

Next, configure the access ports connecting to the hosts and assign them to their respective VLANs.

```
D1(config)# interface GigabitEthernet1/0/6
D1(config-if)# description Access port to PC1
D1(config-if)# switchport mode access
D1(config-if)# switchport access vlan 10
D1(config-if)# exit
D1(config)#
D1(config)# interface GigabitEthernet1/0/18
D1(config-if)# description Access port to PC2
D1(config-if)# switchport mode access
D1(config-if)# switchport access vlan 20
D1(config-if)# exit
```

* switchport mode access
* switchport mode trunk

4. Enable IP routing.

Finally, enable IPv4 routing with the **ip routing** global configuration command to allow traffic to be exchanged between VLANs 10 and 20. This command must be configured to enable inter-VLAN routing on a Layer 3 switch for IPv4.

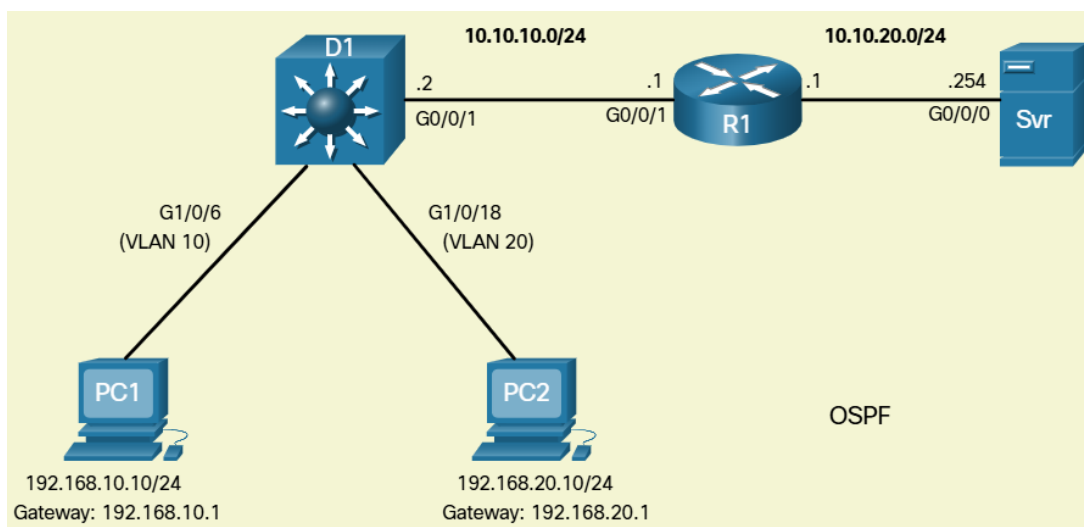
```
D1(config)# ip routing
D1(config)#
```

Routing on a Layer 3 Switch

If VLANs are to be reachable by other Layer 3 devices, then they must be advertised using static or dynamic routing. To enable routing on a Layer 3 switch, a routed port must be configured.

A routed port is created on a Layer 3 switch by disabling the switchport feature on a Layer 2 port that is connected to another Layer 3 device. Specifically, configuring the **no switchport** interface configuration command on a Layer 2 port converts it into a Layer 3 interface. Then the interface can be configured with an IPv4 configuration to connect to a router or another Layer 3 switch.

In the figure, the previously configured D1 Layer 3 switch is now connected to R1. R1 and D1 are both in an Open Shortest Path First (OSPF) routing protocol domain. Assume inter-VLAN has been successfully implemented on D1. The G0/0/1 interface of R1 has also been configured and enabled. Additionally, R1 is using OSPF to advertise its two networks, 10.10.10.0/24 and 10.20.20.0/24.



Complete the following steps to configure D1 to route with R1:

Step 1. Configure the routed port.

Step 2. Enable routing.

Step 3. Configure routing.

Step 4. Verify routing.

Step 5. Verify connectivity.

1. Configure the routed port.

Configure G0/0/1 to be a routed port, assign it an IPv4 address, and enable it.

```
D1(config)# interface GigabitEthernet0/0/1
D1(config-if)# description Routed Port Link to R1
D1(config-if)# no switchport
D1(config-if)# ip address 10.10.10.2 255.255.255.0
D1(config-if)# no shut
D1(config-if)# exit
D1(config)#
```

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Routed port.

2. Enable routing.

Ensure IPv4 routing is enabled with the **ip routing** global configuration command.

```
D1(config)# ip routing
D1(config)#
```

3. Configure routing.

Configure the OSPF routing protocol to advertise the VLAN 10 and VLAN 20 networks, along with the network that is connected to R1. Notice the message informing you that an adjacency has been established with R1.

```
D1(config)# router ospf 10
D1(config-router)# network 192.168.10.0 0.0.0.255 area 0
D1(config-router)# network 192.168.20.0 0.0.0.255 area 0
D1(config-router)# network 10.10.10.0 0.0.0.3 area 0
D1(config-router)# ^Z
D1#
*Sep 17 13:52:51.163: %OSPF-5-ADJCHG: Process 10, Nbr 10.20.20.1 on GigabitEthernet0/0/1 from LOADING to FULL, Loading Done
D1#
```

} Directly connected

4. Verify routing.

Verify the routing table on D1. Notice that D1 now has a route to the 10.20.20.0/24 network.

```
D1# show ip route | begin Gateway
Gateway of last resort is not set
    10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
C       10.10.10.0/30 is directly connected, GigabitEthernet0/0/1
L       10.10.10.2/32 is directly connected, GigabitEthernet0/0/1
O       10.10.20.0/24 [110/2] via 10.10.10.1, 00:00:06, GigabitEthernet0/0/1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, Vlan10
L       192.168.10.1/32 is directly connected, Vlan10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, Vlan20
L       192.168.20.1/32 is directly connected, Vlan20
D1#
```

3✓

5. Verify connectivity.

At this time, PC1 and PC2 are able to ping the server connected to R1.

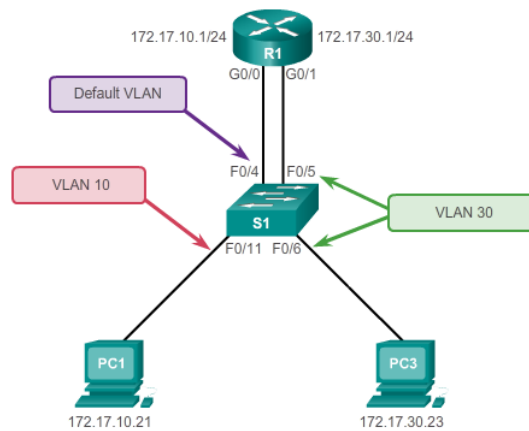
```
C:\Users\PC1> ping 10.20.20.254
Pinging 10.20.20.254 with 32 bytes of data:
Request timed out.
Reply from 10.20.20.254: bytes=32 time<1ms TTL=127
Reply from 10.20.20.254: bytes=32 time<1ms TTL=127
Reply from 10.20.20.254: bytes=32 time<1ms TTL=127
Ping statistics for 10.20.20.254:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss).
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Users\PC1>
!=====
C:\Users\PC2> ping 10.20.20.254
Pinging 10.20.20.254 with 32 bytes of data:
Reply from 10.20.20.254: bytes=32 time<1ms TTL=127
Reply from 10.20.20.254: bytes=32 time<1ms TTL=127
Reply from 10.20.20.254: bytes=32 time<1ms TTL=127
Reply from 10.20.20.254: bytes=32 time<1ms TTL=127
Ping statistics for 10.20.20.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss).
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Users\PC2>
```

Inter-VLAN Configuration Issues

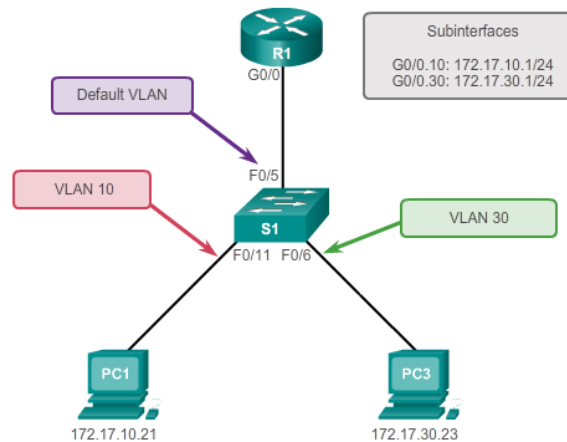
There are several common switch misconfigurations that can arise when configuring routing between multiple VLANs.

When using the traditional routing model for inter-VLAN routing, ensure that the switch ports that connect to the router interfaces are configured with the correct VLANs. If a switch port is not configured for the correct VLAN, devices configured on that VLAN cannot connect to the router interface; therefore, those devices are unable to send data to the other VLANs.

As shown in the Figure below, PC1 and router R1 interface G0/0 are configured to be on the same logical subnet, as indicated by their IP address assignment. However, the switch port F0/4 that connects to router R1 interface G0/0 has not been configured and remains in the default VLAN. Because router R1 is on a different VLAN than PC1, they are unable to communicate. To correct this problem, execute the **switchport access vlan 10** interface configuration mode command on switch port F0/4 on switch S1. When the switch port is configured for the correct VLAN, PC1 can communicate with router R1 interface G0/0, which allows it to access the other VLANs connected to router R1.

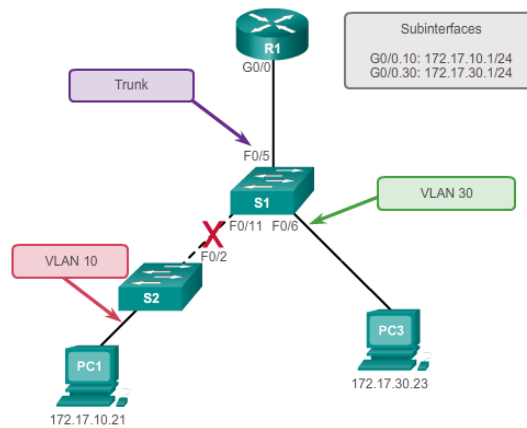


The following Figure topology shows the router-on-a-stick routing model. However, interface F0/5 on switch S1 is not configured as a trunk and is left in the default VLAN for the port. As a result, the router is unable to route between VLANs because each of its configured subinterfaces is unable to send or receive VLAN-tagged traffic.

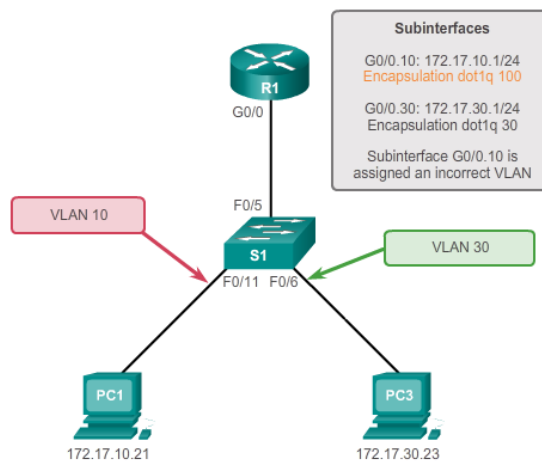


To correct this problem, issue the **switchport mode trunk** interface configuration mode command on switch port F0/5 on S1. This converts the interface to a trunk port, allowing a trunk to be established between R1 and S1. When the trunk is successfully established, devices connected to each of the VLANs are able to communicate with the subinterface assigned to their VLAN, thus enabling inter-VLAN routing.

The Figure topology below shows the trunk link between S1 and S2 is down. Because there is no redundant connection or path between the devices, all devices connected to S2 are unable to reach router R1. As a result, all devices connected to S2 are unable to route to other VLANs through R1. To reduce the risk of a failed inter-switch link disrupting inter-VLAN routing, redundant links and alternate paths should be accounted for within the network design.



With router-on-a-stick configurations, a common problem is assigning the wrong VLAN ID to the subinterface as shown in the Figure below. Router R1 has been configured with the wrong VLAN on subinterface G0/0.10, preventing devices configured on VLAN 10 from communicating with subinterface G0/0.10. This subsequently prevents those devices from being able to send data to other VLANs on the network. Using the **show interface** and the **show running-config** commands can be useful in troubleshooting this issue.



To correct this problem, configure subinterface G0/0.10 to be on the correct VLAN using the **encapsulation dot1q 10** subinterface configuration mode command. When the subinterface has been assigned to the correct VLAN, it is accessible by devices on that VLAN and the router can perform inter-VLAN routing. With proper verification, router configuration problems are quickly addressed, allowing inter-VLAN routing to function properly.

Procedures:

Dear students, please note that the lab problems **sheet**, the **packet tracer activities** and the **practical discussion videos** have been uploaded on your **Microsoft Teams** group. You are required to carefully study this experiment and then complete the lab sheet.

References

Cisco Networking Academy - CCNA: Switching, Routing, and Wireless Essentials.
<https://www.netacad.com>